BRIDGING NEURAL AND BEHAVIORAL DATA WITH XAI IN PREDICTIVE NEUROMARKETING

25-26J-145-DS

Project Proposal Report

Kavindya T A S

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Department of Computer Science

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DECLARATION PAGE

I declare that this is my own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of my knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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The above candidate is carrying out research for the undergraduate Dissertation under my supervision.

Signature of the supervisor: Date

..... 31.08.2025

Mr. Samadhi Rathnayake

ABSTRACT

Comprehending how consumers make decisions in intricate and value-oriented situations poses a significant challenge in the field of neuromarketing research. This investigation aims to model behavioural dynamics to examine how individuals modify their choices when faced with competing influences like sustainability and cost-effectiveness. Controlled experiments for decision-making will be developed using PsychoPy to create realistic scenarios, while behavioural data including eye-tracking trends, response times, and self-reported difficulty will be gathered and prepared for analysis.

To identify hidden decision-making styles, unsupervised learning methods will be utilized to cluster behavioural characteristics, differentiating patterns such as impulsive versus thoughtful strategies. Additionally, sequential learning models like Long Short-Term Memory (LSTM) and Gated Recurrent Units (GRU) will be applied to understand the temporal changes in decision-making processes across various stages. This approach to time-series modelling allows for a nuanced comprehension of how behaviours change in response to pressure, uncertainty, or ethical dilemmas.

The expected results include the discovery of behavioural clusters that illustrate core decision tendencies, alongside predictive models that clarify shifts in decision-making strategies. These results are projected to shed light on how behavioural changes impact consumer preferences and shopping patterns. Ultimately, this research adds to predictive neuromarketing by merging behavioural AI with time-based modelling, providing insights for the development of context-sensitive, interpretable AI systems capable of understanding the complexities of real-world consumer behaviour.

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1. INTRODUCTION

1.1.Background & Literature survey

Behavioural dynamics how people's actions, attention, and choices unfold over time are central to understanding consumer decision-making in real-world contexts. Neuromarketing research has long combined neuroscientific measures (e.g., EEG, fMRI) with behavioural signals (e.g., eye movements, reaction times, self-reports) to go beyond verbal reports and link internal states to choices such as attention, trust, risk perception, and willingness-to-pay. Recent systematic reviews highlight EEG's predictive power for consumer preference and show growing interest in combining EEG with other modalities for improved prediction.

Eye-tracking provides fine-grained temporal and spatial indices of visual attention and has been used both alone and together with neural signals to reveal which features of stimuli drive attention and preference. Multimodal approaches that fuse EEG and eye-tracking have demonstrated improved consumer-choice prediction relative to single-modality models.

Modelling temporal structure is essential because decision strategies can evolve within a trial (e.g., initial impulsive attraction vs. later reflective reappraisal). Recurrent and sequence models such as LSTM and GRU are widely used to capture such temporal dependencies in behavioural and physiological time series; hybrid pipelines that combine clustering with sequence models (e.g., Seq2Seq LSTM after clustering) have shown promise in improving forecasting and personalization in other domains.

Explainable AI (XAI) is an emerging requirement in applied neuromarketing: stakeholders increasingly need interpretable models that reveal why the model predicts a preference or shift in strategy, not only that it does. Recent work applies XAI methods to multimodal neural–behavioural data to extract functional networks or feature.

Finally, experimental platforms such as PsychoPy enable reproducible, time-precise behavioural experiments (stimulus control, reaction-time logging, integration with eye-trackers), making them suitable for the controlled decision scenarios proposed here

1.2.Research Gap

Despite progress, there are still several gaps in modelling dynamic behavioural patterns throughout the stages of decision-making in value-oriented contexts (e.g., sustainability compared to cost).

The integration of dynamic multimodal data at the level of decision stages is lacking. Many neuromarketing investigations combine EEG and eye-tracking data in an aggregated manner (summarized per trial or utilizing static features). A limited number of studies actively track the stage-by-stage temporal dynamics how factors such as attention, reaction time, and neural indicators develop during a decision process and influence shifts between impulsive and reflective decision strategies.

There is a need for behavioural motif segmentation and clustering related to decision-making strategies. Although time-series clustering methods are available, frameworks that focus on behaviour, such as Temporally Aligned Segmentation & Clustering (TASC), are recent and not widely implemented in neuromarketing. Utilizing motif-aware clustering could uncover hidden decision-making styles with a high degree of temporal precision.

The combination of clustering techniques with sequential deep learning models in neuromarketing presents a promising avenue. Hybrid approaches (first clustering, then modelling) have enhanced forecasting in other sectors (e.g., energy management), yet they remain largely underutilized for multimodal consumer decision datasets. There is potential to merge unsupervised strategy-type identification with LSTM/GRU sequence models that forecast transitions and final choices.

The area of explainability for decisions that evolve over time needs further development. Explainable AI (XAI) has been employed with static neural classifiers; however, explainability for sequential, multimodal decision-making models illustrating the moments and reasons behind a participant's movement from impulsive to reflective behaviour amid contextual trade-offs is still in its early stages. This hinders its practical use by marketers and those concerned with ethical considerations.

1.3.Research Problem

The main challenge of this research is to create an explainable, multimodal framework that integrates advanced sequence modelling with behavioral data. Using sequence models such as LSTM and GRU, the framework specifically aims to extract hidden decision-making styles from behavioral time series, model how these styles evolve over time over decision-making stages, and generate interpretable insights into model predictions. Through the use of this methodology, practitioners will be able to understand how behavior changes when cost-efficiency and values like sustainability clash.

Concretely, the project will:

- Design controlled, repeatable decision scenarios using PsychoPy to elicit trade-offs (sustainability vs. cost) while synchronously recording eye-tracking, reaction times, and self-reported difficulty.
- Preprocess and align multimodal time series (eye-gaze x/y, fixation durations, RTs, event markers; EEG band power/ERPs where available), then extract candidate temporal features for analysis.
- Apply temporally-aware segmentation and clustering (e.g., TASC or related motif-clustering) to discover latent decision-making motifs (impulsive, deliberative, vacillating).
- Train sequential neural models (LSTM/GRU, potentially attention-augmented) within and across clusters to predict decision outcomes and stage transitions. Comparative evaluation will determine whether GRU/LSTM or hybrid architectures perform better for this multimodal, individual level data.
- Incorporate XAI methods (feature attribution, temporal saliency maps, attention visualization) to produce interpretable explanations of why the model predicts a choice or a shift in strategy at a given moment.

Together, these steps aim to close the gap between static neuromarketing analyses and actionable, temporally resolved, explainable models of consumer decision dynamics enabling richer behavioral insights and more trustworthy predictive tools for neuromarketing applications.

2. OBJECTIVES

2.1. Main Objectives

To develop an explainable, multimodal framework that identifies latent decision-making styles from behavioral time series, models their temporal evolution across decision stages using sequence models (LSTM/GRU and attention-augmented variants), and provides interpretable explanations of model predictions to reveal how behavioral shifts occur in value-laden choice scenarios (e.g., sustainability vs. cost-efficiency).

2.2. Specific Objectives

1. **Design controlled decision-making experiments.** Create repeatable PsychoPy paradigms that present value trade-offs (sustainability vs. cost) and record synchronized behavioral signals (eye-tracking, reaction times, self-reports, and event markers).
2. **Collect a multimodal dataset.** Recruit a representative sample, capture high-resolution eye-tracking and timestamped behavioral logs, and (optionally) collect concurrent neural signals (EEG) if available; document metadata and ethical procedures.
3. **Preprocess and align time series.** Implement pipelines for cleaning, synchronizing, and normalizing multimodal streams (interpolation of gaze, fixation detection, RT cleaning, epoching EEG, artifact rejection) and produce feature sets suitable for temporal modeling.
4. **Extract temporal behavioral features.** Derive stage-level and fine-grained features (fixation durations, saccade metrics, gaze transition matrices, time-to-first-fixation, RT curvature, temporal windows of bandpower/ERP features) that capture decision dynamics.
5. **Discover latent decision-making styles.** Apply temporally aware clustering/segmentation methods (e.g., motif clustering, TASC, autoencoder-based clustering) to identify recurring strategy types (impulsive, reflective, vacillating) and evaluate cluster quality (silhouette, Davies–Bouldin, NMI against any labeled probes).
6. **Develop sequential predictive models.** Train and compare sequence models (LSTM, GRU, attention-augmented LSTM/Transformer hybrids) to (a) predict final choices, and (b) predict within-trial transitions between decision stages, using both cluster-conditioned and global modeling approaches.
7. **Integrate XAI methods.** Incorporate explainability techniques appropriate for sequential multimodal models (temporal saliency maps, Integrated Gradients, attention visualization, SHAP for time series) to produce human-interpretable explanations of why a model predicts a particular choice or strategy shift.
8. **Evaluate performance and interpretability.** Quantitatively evaluate predictive accuracy (accuracy, F1, AUC), temporal prediction quality (precision/recall for stage transitions), clustering validity, and interpretability metrics (fidelity,

stability, user-centered interpretability checks), plus statistical tests for behavioral differences across scenarios.

9. **Compare modeling strategies and generalization.** Compare LSTM vs. GRU vs. attention/hybrid architectures and test model generalizability across participant subgroups and scenario types (e.g., different levels of cost vs. sustainability trade-offs).
10. **Deliver actionable insights and guidelines.** Produce a set of practical recommendations and visualization tools for practitioners (marketers, UX researchers) on interpreting temporal behavioral signatures and applying explainable predictive models in neuromarketing contexts.

3. METHODOLOGY

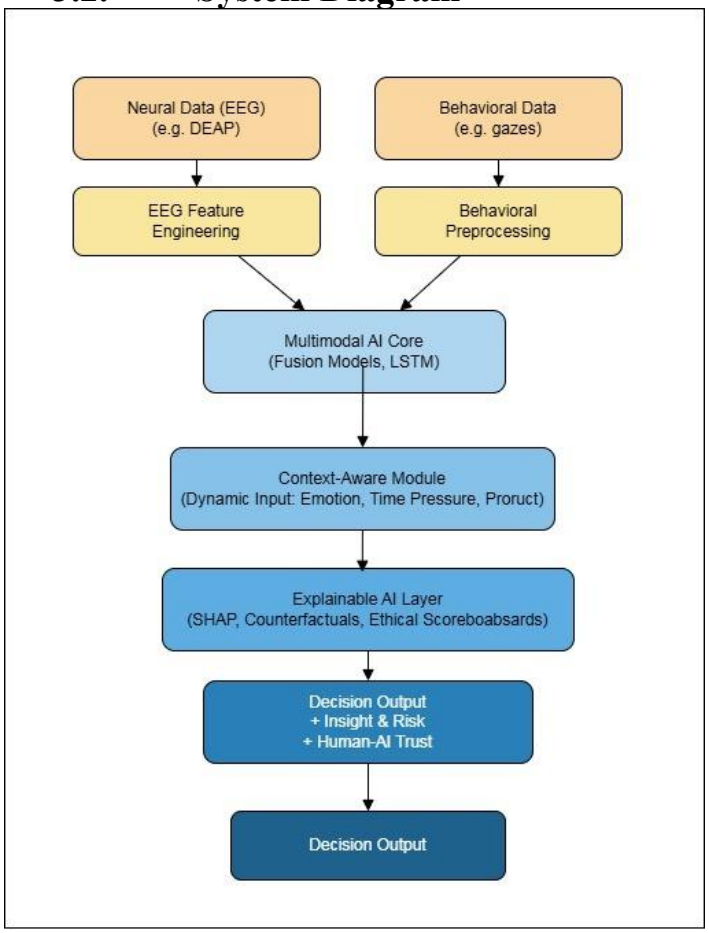
3.1. System Overview

An explainable, multimodal behavioral modelling framework for examining the dynamics of consumer decision-making is developed by the suggested system. PsychoPy will be used to construct controlled experiments in which participants make trade-off judgements (e.g. sustainability vs. cost-efficiency). Eye tracking, response times, and self-reported choice difficulties are among the behavioral data that will be gathered throughout these tasks.

To eliminate noise, synchronize streams, and extract significant temporal patterns, the raw behavioral data will undergo preprocessing. Unsupervised clustering will then be used to find latent decision-making styles, such as reflective and impulsive. The temporal evolution of behaviors across decision phases will then be captured by sequential deep learning models (LSTM/GRU). Finally, to interpret predictions and offer practical insights into how and why decisions change under contextual trade-offs, Explainable AI (XAI) techniques such as saliency maps, SHAP, and attention visualization will be used.

In order to provide a context-aware, interpretable predictive system for neuromarketing, this methodology combines data-driven modelling with experimental behavioral research.

3.2. System Diagram



3.3. Data Collection

Data Needed-

- Eye tracking metrics (fixation duration, saccades, gaze transitions)
- Reaction times (RTs per decision stage)
- Self-reported difficulty ratings after each trial

Collection Tools- PsychoPy integrated with eye-tracking devices, digital survey forms for self-reports.

Participants- Volunteer participants representing diverse consumer profiles

4. PROJRCT REQUIREMENT

4.1. Functional Requirement

- The system should be able to collect and integrate multimodal data (behavioral time series, textual feedback, contextual attributes)
- The system should identify latent decision-making patterns using clustering or representation learning
- The system should model temporal evolution of decision-making patterns using sequence models (LSTM/GRU)
- The system should generate explanations of predictions using explainable AI techniques. (SHAP, LIME, attention visualization)
- The system should provide an interactive dashboard for users to explore results.

4.2. Non-Functional Requirement

- Performance – The system should handle medium scale datasets efficiently
- Scalability – The architecture should support future integration of larger datasets and additional modalities.
- Usability- The dashboard should be intuitive, supporting both technical and non-technical users.
- Security – Data must be anonymized and comply with data protection standards.
- Reliability – The framework should give stable, reproducible results.

4.3. User Requirements

- Researchers/Data Scientists – Need to test hypotheses about decision making behavior and interpret results.
- Practitioners
- Educators/ Trainers

4.4. System Requirements

Hardware

- Minimum – 16 GB RAM, 6-core CPU, 4GB GPU

- Recommended – 32GB RAM, NVDIA GPU(8GB+)

Software

- Python (TensorFlow/PyTorch, Scikit-learn).
- Explainability libraries (SHAP, LIME, Captum)
- Database – PostgreSQL / MongoDB for multimodal data storage
- Dashboard – Streamlit, Dash, PowerBI

4.5. Use Cases

1. Identify decision style
   - Actor – Researcher
   - Description – Uploads behavioral dataset, system detects latent decision-making style
2. Temporal evolution analysis
   - Actor – Researcher
   - Description – Runs LSTM/GRU model to track decision shifts over time.
3. Explain model predictions
   - Actor – Practitioner
   - Description – Selects a decision stage, system outputs SHAP/LIME explanations
4. Visualizations
   - Actor – Any user
   - Description – Explore behavioral trajectories via dashboard charts.

4.6. Test Cases

- Uploads multimodal datasets- System integrates and preprocesses correctly.
- Run clustering on behavioral data – system outputs distinct decision-making groups.
- Train sequence model (LSTM/GRU)- model >= target accuracy.
- Explainability module – explanations are generated within acceptable response time
- Dashboard – all visualizations render without errors.

5. Description of Personal and Facilities

5.1. Personal

Using PsychoPy, the researcher will create and carry out controlled decision-making scenarios (such as cost-effectiveness versus sustainability). Among the duties are gathering and preprocessing behavioral data, such as judgements of decision difficulty, eye-tracking patterns, and reaction times. The researcher will create sequential models (LSTM/GRU) to capture behavioral dynamics and use unsupervised learning approaches to determine decision-making styles.

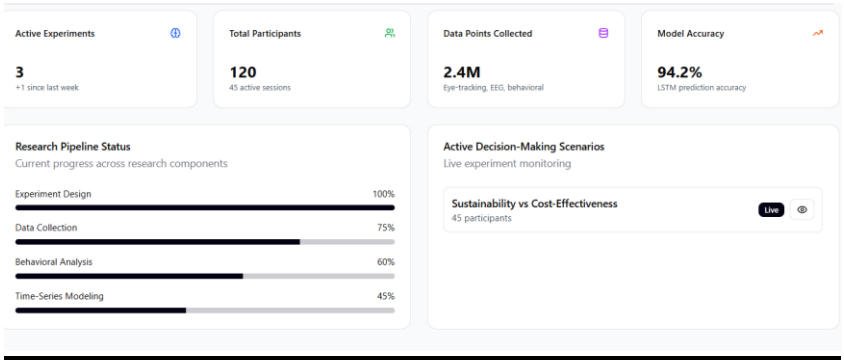
5.2. Facilities

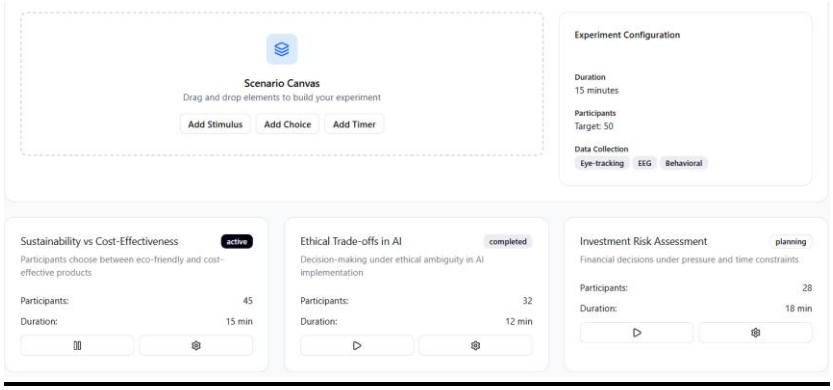
A personal computer using Python-based tools (PsychoPy, TensorFlow/PyTorch, scikit-learn, and pandas) will be used to create this component. PsychoPy will be used to design the behavioral experiment, and Jupyter Notebook or comparable software will be used for time-series modelling and clustering. If hardware is insufficient, eye-tracking data will be gathered using simulated datasets or accessible lab resources. This component will also be supported by computing resources, supervisor supervision, and university digital libraries.

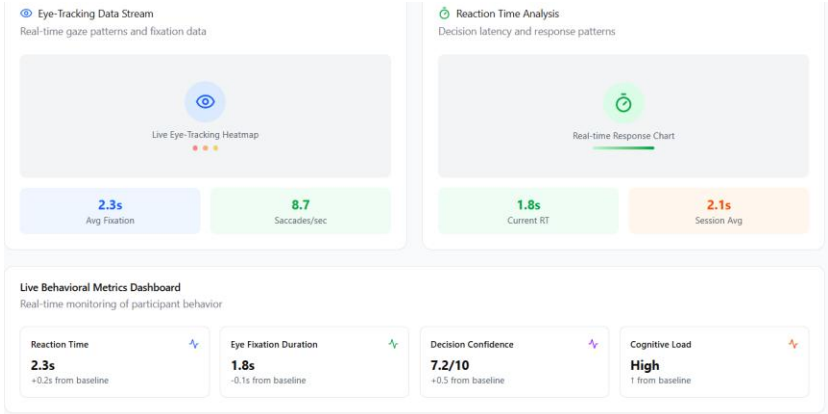
6. Grant Chart

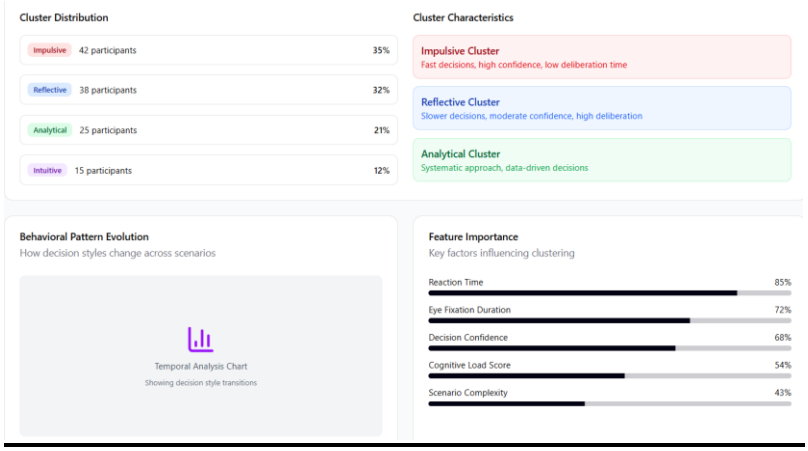
ID	Task	2025-2026																							
		June		July		August		September		October		November		December		January		February		March		April		May	
1	Topic Selection																								
2	Topic Assesment																								
3	Literature Review																								
4	TAF																								
5	TAF Evaluation																								
6	Charter																								
7	Proposal Draft																								
8	Proposal presentation																								
9	Propsal Report																								
10	System Development																								
11	Progress Presentation 01																								
12	Reserch Paper																								
13	Progress Presentation 02																								
14	Final Report Submission																								
15	Final Report Feedback																								
16	Final Presentation and Viva																								

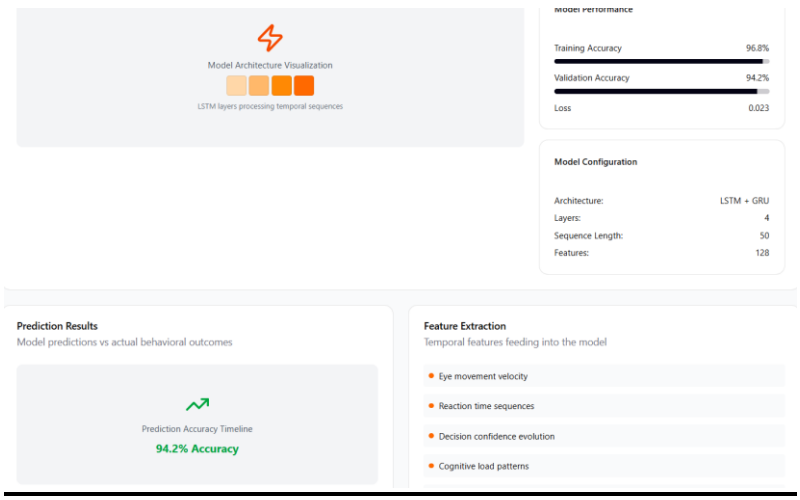
7. WIREFRAMES

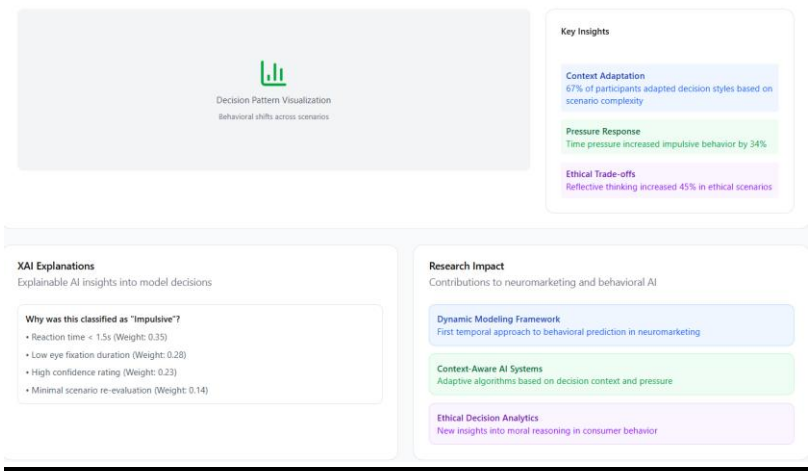







Cluster Distribution
Impulsive 42 participants 35%
Reflective 38 participants 32%
Analytical 25 participants 21%
Intuitive 15 participants 12%
Cluster Characteristics
Impulsive Cluster
Fast decisions, high confidence, low deliberation time
Reflective Cluster
Slower decisions, moderate confidence, high deliberation
Analytical Cluster
Systematic approach, data-driven decisions
Behavioral Pattern Evolution
How decision styles change across scenarios
Temporal Analysis Chart
Showing decision style transitions
Feature Importance
Key factors influencing clustering
Reaction Time 85%
Eye Fixation Duration 72%
Decision Confidence 68%
Cognitive Load Score 54%
Scenario Complexity 43%


Model Architecture Visualization
LSTM layers processing temporal sequences
Training Accuracy 96.8%
Validation Accuracy 94.2%
Loss 0.023
Model Configuration
Architecture: LSTM + GRU
Layers: 4
Sequence Length: 50
Features: 128
Prediction Results
Model predictions vs actual behavioral outcomes
Prediction Accuracy Timeline
94.2% Accuracy
Feature Extraction
Temporal features feeding into the model
Eye movement velocity
Reaction time sequences
Decision confidence evolution
Cognitive load patterns


Decision Pattern Visualization
Behavioral shifts across scenarios
Key Insights
Context Adaptation
67% of participants adapted decision styles based on scenario complexity
Pressure Response
Time pressure increased impulsive behavior by 34%
Ethical Trade-offs
Reflective thinking increased 45% in ethical scenarios
XAI Explanations
Explainable AI insights into model decisions
Why was this classified as "Impulsive"?
• Reaction time < 1.5s (Weight: 0.35)
• Low eye fixation duration (Weight: 0.28)
• High confidence rating (Weight: 0.23)
• Minimal scenario re-evaluation (Weight: 0.14)
Research Impact
Contributions to neuromarketing and behavioral AI
Dynamic Modeling Framework
First temporal approach to behavioral prediction in neuromarketing
Context-Aware AI Systems
Adaptive algorithms based on decision context and pressure
Ethical Decision Analytics
New insights into moral reasoning in consumer behavior

8. REFERENCE

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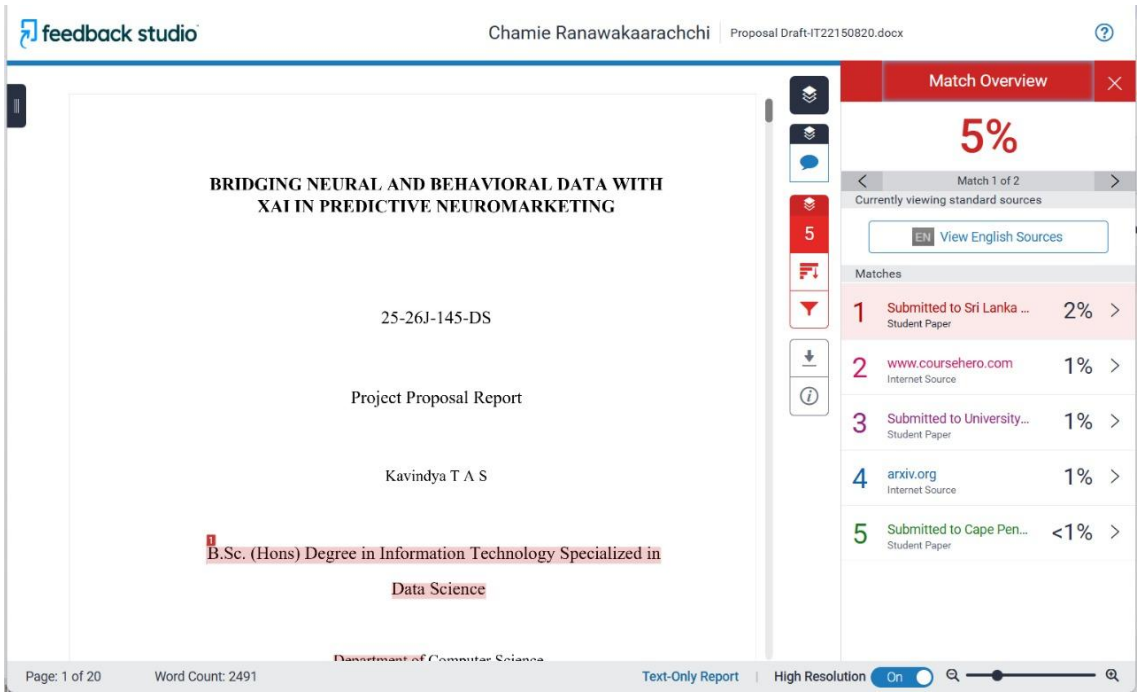
Plagiarism Report